

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

2nd November 2015

Subject CT7 – Business Economics

Time allowed: Three Hours (10.30 to 13.30 Hrs.)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *Mark allocations are shown in brackets.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet. However, answers to objective type questions could be written on the same sheet.*
4. *Please check if you have received complete Question Paper and no page is missing. If so kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

Q. 1) The economic problem of HOW a good shall be produced (the capital-labour mix) is solved in capitalist societies:

- A. Through the purchasing decisions made by consumers in the market place for the final good produced
- B. By means of the profit motive which prompts producers to keep their production costs to a minimum
- C. Purely by comparing the cost of capital relative to the cost of labour.
- D. By the use of relatively capital intensive methods of production

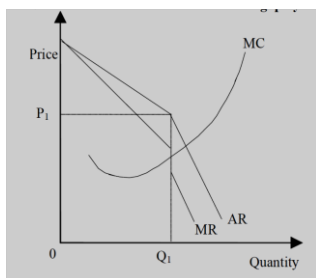
[1.5]

Q. 2) Which of the following is NOT an impact of introducing a price ceiling - set below the free market price

- A. Price will be reduced
- B. Quantity supplied will fall
- C. Quantity demanded will rise
- D. There will be surplus

[1.5]

Q. 3) In the following diagram, Demand curve is kinked at the current price, it is because



- A. Demand curve is more elastic above the current price and more inelastic below
- B. Demand curve is more inelastic above the current price and more elastic below
- C. Demand curve is less elastic above the current price and less inelastic below
- D. Demand curve is less inelastic above the current price and less elastic below

[1.5]

Q. 4) What among the following corresponds to vertical demand curve?

- A. $PED = 0$
- B. $PED = -1$
- C. $PED = 1$
- D. $PED = \infty$

[1.5]

Q. 5) Assuming the good is not a Giffen good, in which of the following case will the total revenue remain unchanged due to x% fall in the price

- A. $PED = 0$
- B. $PED = -1$
- C. $PED = 1$
- D. $PED = \infty$

[1.5]

- Q. 6)** Assuming the good is not a Giffen good, the total revenue from sale of goods will rise if
- A.** Price rises and demand for the good is price-elastic
 - B.** Price rises and demand for the good is price-inelastic
 - C.** Consumer income rises and the good is inferior
 - D.** Consumer income falls and the good is a normal good
- [1.5]**
- Q. 7)** The kinked demand curve model of oligopoly is based upon the assumption that:
- A.** A firm's competitors match both its price increases and price reductions
 - B.** One firm in the industry sets the price for all other firms
 - C.** A firm's competitors match its price reductions but not its price increases
 - D.** The price charged by the firm can either rise or fall depending on what happens to its competitors' prices
- [1.5]**
- Q. 8)** A price taker firm has:
- A.** Upward sloping demand curve
 - B.** Downward sloping demand curve
 - C.** Horizontal demand curve
 - D.** Vertical demand curve
- [1.5]**
- Q. 9)** A Taxi service charges higher price during 5 pm to 11 am. This is an example of
- A.** Variable pricing
 - B.** Peak load pricing
 - C.** Inter temporal pricing
 - D.** Two part tariff
- [1.5]**
- Q. 10)** The prisoner's dilemma, applied to a situation involving two oligopolists, illustrates that:
- A.** Each firm will not take account of its rival's reactions when making its decision.
 - B.** The price set by one firm will not influence the price of the other firm.
 - C.** In avoiding the worst possible outcome the firms will fail to reach the best possible outcome.
 - D.** In avoiding the worst possible outcome the firms will succeed in reaching the best possible outcome.
- [1.5]**
- Q. 11)** Following is not a feature of a Perfect Competition
- A.** Firms face perfectly inelastic demand curve
 - B.** Firms produce identical product
 - C.** Firms face no entry barriers
 - D.** Firms are price takers
- [1.5]**

- Q. 12)** Which one of the following is NOT a barrier to entry into a monopoly market?
- A. Control over key inputs
 - B. Heavy potential advertising costs.
 - C. Large capital requirements.
 - D. Constant returns to scale. [1.5]
- Q. 13)** Constant returns to scale means that as a firm's scale of production is increased:
- A. Long run total costs remain constant
 - B. Total output remains unchanged
 - C. Long run average costs and long run marginal costs are equal
 - D. Fixed costs remain constant [1.5]
- Q. 14)** Which of the following causes deadweight welfare loss
- A. Externalities
 - B. Taxes or subsidies
 - C. Binding price ceilings or floors
 - D. All of the above [1.5]
- Q. 15)** Which of the following can be used by the government to encourage firms to undertake research and development
- A. Patent system
 - B. Diffusion policies
 - C. Cooperative R&D
 - D. All of the above [1.5]
- Q. 16)** The table below shows the quantities of computers and mobile phones that one unit of input can produce in India and China. Assume that these are the only two goods that the countries produce.

	One unit of input produces:	
	Computers	Mobile Phones
India	20	200
China	40	300

Given the table above, which of the following is true?

- A. China is likely to export Mobile Phones to India.
- B. India has an absolute advantage in the production of Mobile Phones.
- C. China has a comparative advantage in the production of Mobile Phones.
- D. China has an absolute advantage in the production of Mobile Phones. [1.5]

- Q. 17)** Which of the following is example of Interventionist supply-side policies
- A.** A reduction in the size of the public sector and promotion of the private sector
 - B.** Provide investment grants to promote R&D
 - C.** The introduction of new legislation to reduce the power of the trade union
 - D.** Improvements in infrastructure such as better roads
- [1.5]**
- Q. 18)** Which of the following rules are NOT followed by WTO members
- A.** Provide trade concession to all members if extended to one of the member
 - B.** Provide consistency with regards to taxation and subsidies
 - C.** No alteration to tariffs without negotiation with their trading partners
 - D.** The prohibition of quotas
- [1.5]**
- Q. 19)** Given the following data on simple closed economy:
- $C = 10 + 0.75Y$
 $I = 20$
 $G = 40$
- where C is consumer expenditure, Y is national income, G is government expenditure on goods and services and I is investment expenditure.
- Calculate the new level of national income if government expenditure increased by 10 million?
- A.** 310
 - B.** 320
 - C.** 280
 - D.** 300
- [1.5]**
- Q. 20)** Which of the following is possible non-economic reasons to restrict trade
- A.** To prevent the importation of harmful goods
 - B.** To reduce the influence of trade on consumer tastes
 - C.** To impose trade sanctions on countries with which it disagrees politically
 - D.** To prevent “dumping” and other unfair trade practices
- [1.5]**
- Q. 21)** Which one of the following is an advantage for a country adopting a flexible exchange rate system regime?
- A.** It provides certainty for organisations engaged in international trade
 - B.** It eliminates transaction costs
 - C.** Monetary policy can be used to manage exchange rate
 - D.** It reduces the need for central banks to keep reserves of foreign exchange
- [1.5]**

- Q. 22)** Under floating exchange rates:
- A.** Domestic inflation rates are unavoidably linked across national boundaries
 - B.** Domestic inflation in one country can lead to inflation in another even if the exchange rate moves to preserve purchasing power parity
 - C.** Domestic inflation is dictated outside the bounds that would have constrained prices in a fixed rate regime
 - D.** None of the above
- [1.5]**
- Q. 23)** Structural unemployment is unemployment that:
- A.** Increases in a recession and falls in a boom
 - B.** Arises when the unemployed lack the skills needed by newly created jobs
 - C.** Arises when those seeking work give up hope of finding a job
 - D.** Occurs as the result of a transition from one job to another
- [1.5]**
- Q. 24)** In latest budget speech, top rate of income tax is cut from 30% to 25%. Other things remaining the same, the effect on the multiplier will be:
- A.** A rise
 - B.** A fall
 - C.** No change
 - D.** Uncertain
- [1.5]**
- Q. 25)** If the money supply decreases due to a contractionary open market operation by the central bank then the price of treasury bills will:
- A.** Fall as the short term interest rate rises
 - B.** Fall as the short term interest rate falls
 - C.** Rise as the short term interest rate rises
 - D.** Rise as the short term interest rate falls
- [1.5]**
- Q. 26)** If the government imposed a price for a good that was above the equilibrium price, the consequence would be:
- A.** A contraction of demand, an increase in supply and a market surplus
 - B.** A decrease in demand, an extension of supply and a market surplus
 - C.** A contraction in demand, an extension in supply and a market surplus
 - D.** A rise in supply, a fall in demand and a market shortage
- [1.5]**
- Q. 27)** Which one of the following is not a benefit to a country from joining a single currency area?
- A.** Reduced transaction costs
 - B.** Lower interest rate
 - C.** Reduced exchange rate uncertainty
 - D.** Increased price transparency
- [1.5]**

- Q. 28)** The economic welfare case for governments increasing taxes on petrol to raise its real price is that:
- A.** Oil is a scarce resource.
 - B.** It would reduce the imports of oil.
 - C.** There is a large demand for petrol.
 - D.** Petrol consumption involves external social costs. **[1.5]**
- Q. 29)** Which of the following will lead to the closure of a profit maximizing firm in the short run
- I.** Total revenue is less than total variable cost
 - II.** Total cost is greater than total revenue
 - III.** Fixed costs are greater than total revenue
- A.** I and II
 - B.** II and III
 - C.** I only
 - D.** III only **[1.5]**
- Q. 30)** Suppose that the demand function in a market is given by $Q=100 - 2P$ (where Q is quantity demanded and P is price) and you are the only seller in this market. Your total revenue and average revenue as functions of output are as follows:
- A.** Total Revenue = $50Q-(Q^2/2)$; Average Revenue = $50-(Q/2)$
 - B.** Total Revenue = $100Q-(Q^2/2)$; Average Revenue = $100-(Q/2)$
 - C.** Total Revenue = $50Q^2-(Q^2/2)$; Average Revenue = $50Q-(Q/2)$
 - D.** Total Revenue = $100Q-(Q^2/2)$; Average Revenue = $100Q-(Q/2)$ **[1.5]**
- Q. 31)** Define the following terms:
- i)** Supernormal profit
 - ii)** Short-run shut-down point
 - iii)** Minimum efficient scale
 - iv)** Market clearing
 - v)** Free market
 - vi)** Nash Equilibrium
 - vii)** Valuation ratio **[3.5]**
- Q. 32)** Explain what insurance company can do to reduce the problems of adverse selection and moral hazard? **[2]**
- Q. 33)** What is the difference between stabilizing and destabilizing speculation? Explain with the help of an example and diagram. **[6]**

- Q. 34)** A firm with market power faces an inverse demand curve for its product of $P = 10 - Q$. Assume that the firm faces a marginal cost curve of $MC = 1 + Q$.
- i)** If the firm cannot price discriminate, what is the profit maximizing level of output and price? (2)
 - ii)** If the firm cannot price discriminate, what are the levels of consumer and producer surplus in the market, assuming the firm maximizes its profit? Calculate the deadweight loss from the market power. (5)
 - iii)** If the firm has the ability to practice perfect price discrimination, what is the firm's output? (1)
 - iv)** If the firm practices perfect price discrimination, what are the levels of consumer and producer surplus? What is the deadweight loss from the market power? (1.5)
[9.5]
- Q. 35)** What are the possible motives for growth via vertical integration? What are the problems associated with vertical integration? [3]
- Q. 36)** Describe the intended effects of advertising [4]
- Q. 37)** Discuss the effectiveness of supply side policies. [4]
- Q. 38)** Explain the accelerator principle and the influence the accelerator has on the operation of the business cycle. [5]
- Q. 39)** Explain why monetarists feel that in the long run Philips Curve is vertical. [3]
- Q. 40)** Identify and explain which of the following economic events with all other things being equal results in cost-push inflation and which will result in demand-pull inflation.
- i)** An increase in government spending
 - ii)** An increase in corporate profit margins
 - iii)** An increase in nominal wages
 - iv)** A decrease in the price of raw material [4]

Q. 41) Given the following data on Country A's balance of payments:

Exports of goods + 200mn
Imports of goods -150mn
Exports of services +120mn
Imports of services -160mn
Interest, profits and dividends received +15mn
Interest, profits and dividends paid -10mn
Unilateral receipts +30mn
Unilateral payments -20mn
Capital outflows -185mn
Capital inflows +140mn
Statistical error?
Change in reserves [rise (-), fall (+)] +10mn

i) Calculate balance on trade in goods and services

ii) Calculate the "invisibles" balance.

iii) Calculate the current account balance.

iv) Calculate the balance for official financing.

v) Determine the statistical error.

[5]

Q. 42) Central bank generally balances the conflicting aim of banks between liquidity and profitability. Please list the other method by which banks may manage this conflict along with advantages and disadvantages of the method.

[6]
